APPENDIX
DEFINITIONS OF FAST REACTOR BREEDING AND INVENTORY

THIS note sets out the preferred definitions in the U.K. concerned with plutonium utilization in fast reactors.

Plutonium

1. Since the plutonium inventory of a fast reactor may arise from sources of plutonium of differing isotopic composition, it is convenient to express a quantity of plutonium as the quantity of Pu 239 which has the same reactivity worth in fast reactors, referred to as "equivalent Pu 239" ($\equiv 239$). Typically, for a large ceramic fuelled fast reactor,

$$\equiv 239 = Pu_{239} + 1{\cdot}5\ Pu_{241} + 0{\cdot}15\ (Pu_{240} + Pu_{242}).$$

2. An alternative, though somewhat less logical, definition of plutonium is

$$\text{Thermally fissile Pu} = Pu_{239} + Pu_{241}.$$

Breeding

3. The preferred measure of breeding performance is

Breeding gain = excess $\equiv 239$ produced per fission above that required to maintain reactivity.

An alternative parameter is

Breeding ratio = total thermally fissile Pu produced per unit of thermally fissile Pu destroyed.

4. If equivalent Pu 239 is identified with thermally fissile plutonium and if an atom of thermally fissile plutonium is destroyed for every heavy atom fissioned, then

Breeding gain = (breeding ratio) − 1.

Inventory

5. The preferred U.K. definition is

Plutonium inventory = $\equiv 239$ in (core + fresh fuel hold-up
+ spent fuel hold-up
+ reprocessing and fabrication hold-up).

Each hold-up is taken over a period of 3 months, that for reprocessing and fabrication being based on fresh fuel. The inventory thus refers to a specified load factor, increasing with it. No allowance is included for commissioning of the station or its approach to equilibrium.

6. Inventory, as defined above, may be expressed in terms of station output as either

Specific inventory = plutonium inventory per unit of net thermal (or electrical) output

or

Inventory rating = (specific inventory)$^{-1}$.

Doubling Time

7. The preferred U.K. definition is

Linear doubling time = time to attain one plutonium inventory by plutonium production from one fast reactor at fuel cycle equilibrium and at specified load factor.

8. An alternative definition is

Exponential doubling time = time in which fast reactor capacity is doubled if it is continuously extended by the immediate addition to its inventory of the plutonium it produces. (Fuel cycle equilibrium and a specified load factor are again implied.)

It follows that

Exponential doubling time = (linear doubling time) $\times$ 0·693.

Choice of Parameters

9. If the parameters defined above are grouped into three columns:

Breeding gain	Pu inventory	Linear doubling time
	Specific inventory	Exponential doubling time
Breeding ratio	Inventory rating	

then selection of one parameter from one column and another from one of the other two columns is sufficient to determine the remaining parameters.